

Catherine Zhao

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EDUCATION

Massachusetts Institute of Technology (MIT) | *B.S. in Computer Science* Expected May 2029

- **Relevant Coursework:** Linear Algebra & Optimization; Algorithms; Machine Learning; C & Low-Level Assembly; Shape Analysis (grad); Deep Learning (grad); Computation Structures; Probability and Random Variables

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, TypeScript, HTML/CSS, Julia, Assembly, SQL

Tools/Frameworks: Git/GitHub, React, Flask, VS Code, PlatformIO, Vite, ITK-Snap, Blender, Perforce

Systems & HPC: Unix, Bash, HPC cluster workflows, SSH/SSHFS, SLURM

WORK EXPERIENCE & RESEARCH

InterSystems, Boston, MA - *Application Services Intern* May 2026 - Aug. 2026

- Developed custom Jira project and CCR type to manage hotfix tracking across TrakCare editions
- Built integration logic between Jira, CCR, and Perforce, enabling synchronized workflow transitions, automated file retrieval and change control, and streamlined code deployment and migration across branches.

Medical Vision Group, MIT CSAIL - *Undergraduate Researcher* Jan. 2026-May 2026

- Developed automated pipeline for reconstruction of 3D placenta shape from ~36k MRI scan volumes via marching cubes, PCA alignment, convex hull fitting, and SDF analysis to support maternal-health research.

PROJECTS

ScribbleHero, MIT Reality Hackathon (Team of 4) Jan. 2026

- Developed *ScribbleHero*, an AR drawing puzzle game built for Snap Spectacles using Snap Lens Studio and Google Gemini. Won honorable mention in the Game Prototyping Track.
- Implemented event-driven systems in Typescript to manage user interaction and physics-based animations.

Tactile 3D Viewer, Assistive Technology Club (Team of 5) Oct. 2025 - Jan.2026

- Built a full-stack web tool allowing legally blind users to explore 3D CAD models via 2D tactile slices.
- Used Trimesh for ray-mesh intersection and slicing, with caching for low-latency rendering.

LEADERSHIP & ENGAGEMENT

MIT Assistive Technology Club - *Project Manager* Feb. 2026-Present

- Organized Git & software principles training workshop, helped plan study breaks and club-wide events, designed promotional content for club outreach, represented club at the Boston Museum of Science Disability Pride Event.

MIT Asian Dance Team - *Choreographer, Cover Director (prev), Promo Maker (prev)* Mar. 2026-Present

- Co-led a team of 8 dancers to perform and film a dance cover for the MIT ADT YouTube channel; taught choreography, oversaw editing, designed promotional content.

MIT Admissions - *Tour Guide* Feb. 2026-Present

- Explained research, academic, and extracurricular opportunities at MIT to groups of ~30 campus visitors.

Harvard Square Homeless Shelter - *Volunteer* Feb. 2026-Present

- Prepared weekly food, clothing, and toiletry orders for ~30 shelter guests and community members experiencing homelessness.